Q1.

The table shows the observations made when four metals are added to cold water and to dilute hydrochloric acid.

metal	observations with cold water	observations with dilute hydrochloric acid
zinc	no reaction	bubbles of gas form and the metal slowly dissolves
platinum	no reaction	no reaction
potassium	the metal floats and then melts, a flame appears, and sometimes there is an explosion	(cannot be done safely)
nickel	no reaction	a few bubbles of gas form if the acid is warmed

(a) Write the names of these **four** metals in the order of their reactivity.

most reactive .....

.....

least reactive .....

.....

1 mark

(b) (i) Give the name of another metal, **not** in the table, which reacts in a similar way to potassium.

.....

1 mark

(ii) What gas is formed when zinc reacts with dilute hydrochloric acid?

.....

1 mark

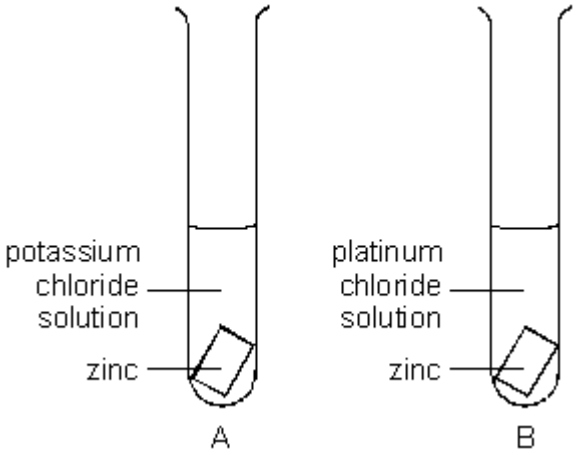
(iii) The experiment with potassium and dilute hydrochloric acid should **not** be done in school laboratories. Suggest why it is dangerous.

.....

.....

1 mark

(c) A scientist set up two test-tubes as shown below.



In test-tube B the zinc strip was slowly covered with a grey deposit.
Nothing happened in the other test-tube.

(i) What was the grey deposit in test-tube B?

.....

1 mark

(ii) Why was this grey deposit formed in test-tube B?

.....

1 mark

(iii) Explain why **no** reaction took place in test-tube A.

.....

.....

1 mark

Maximum 7 marks

Q1.

(a)
- potassium
- zinc
- nickel
- platinum

1 (L5)

all four *metals in the correct order are required for the mark*

(b) (i) sodium

accept 'caesium'

1 (L6)

(ii) hydrogen

1 (L6)

(iii) any **one** from

- the reaction is too violent
  *accept 'it explodes' **or** 'acid goes everywhere'*
- potassium is too reactive

1 (L5)

(c) (i) platinum

1 (L6)

(ii) **answers must refer to the reactivity of both metals**

any **one** from

- zinc displaces platinum from solution
- zinc is higher than platinum in the reactivity series
  accept 'zinc is more reactive than platinum'
- platinum is lower than zinc in the reactivity series
  accept 'platinum is less reactive than zinc'

1 (L6)

(iii) any **one** from

- zinc is less reactive than potassium
- potassium is more reactive than zinc
  *accept 'zinc does not displace potassium from potassium chloride **or** the solution'*

1 (L6)

[7]